Experiment No. 4

4. Develop a LaTeX script to create the Certificate Page of the Report /* Save file as pg4*/

```
\documentclass[a4paper,12pt]{article}
\usepackage{graphicx}
\usepackage{ragged2e}
\usepackage{xcolor}
\usepackage[margin=1in]{geometry}

\begin{document}

	\pagenumbering{roman}

	\begin{center}
		\textbf{\Large Visvesvaraya Technological University}\\
		\textbf{\small Jnana Sangama, Belagavi}

		\vspace{5mm}

		\includegraphics[scale=0.5]{Desktop/vtu logo.png}

		\vspace{5mm}

		\textbf{\small DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING}

		\vspace{5mm}

		\textbf{\Large CERTIFICATE}
	\end{center}

	\vspace{10mm}

	\justify
	This is to certify that Mr. Suresh bearing USN: 1AH20CS001 is a bonafide student of
	Bachelor of Engineering course of the Department of Computer Science and Engineering,
```

ACS College of Engineering, Bangalore, affiliated to Visvesvaraya Technological University,
Belagavi. The project report entitled ``BLOCKCHAIN TECHNOLOGY" is prepared by him
under the guidance of Mr. Charan M S in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering of Visvesvaraya Technological University, Belagavi, Karnataka.

\vspace{15mm}

BCSL456D – LaTeX Laboratory

IV SEM, BE

Dept. of CSE, ACSCE, Bangalore – 560074 \hfill Page No: 11

\vspace{20mm}

.... \hspace{15mm}
.... \hspace{15mm}
....

\vspace{5mm}

Mr. Charan M S \hspace{15mm}
Dr. Senthil Kumaran T \hspace{15mm}
Dr. Ananda Gurdi

\vspace{5mm}

Signature of Guide \hspace{15mm}
Signature of HoD \hspace{15mm}
Signature of Principal

\vspace{15mm}

\begin{center}
\textbf{\small EXTERNAL EXAMINER}
\end{center}

\vspace{10mm}

Name of Examiners \hspace{60mm} Signature with date

\vspace{10mm}

1.

\vspace{10mm}

2.

\end{document}

Experiment No. 5

5. Develop a LaTeX script to create a document that contains the following table with proper labels

```
\documentclass[a4paper,12pt]{article}
\usepackage{graphicx}
\usepackage{caption}

\begin{document}
\begin{center}
\textbf{\Large Student Information Table}
\end{center}

\begin{table}[h]
\centering
\caption{Student Details}
\label{tab:student}

\begin{tabular}{|c|l|c|c|}
\hline
Sl.No & Name & USN & Semester \\
\hline
1 & Suresh & 1AH20CS001 & IV \\
\hline
2 & Ramesh & 1AH20CS002 & IV \\
\hline
3 & Mahesh & 1AH20CS003 & IV \\
\hline
4 & Ganesh & 1AH20CS004 & IV \\
\hline
\end{tabular}

\end{table}

Reference of Table: Table \ref{tab:student}

\end{document}
```

Experiment No. 6

6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

```
\documentclass[a4paper,12pt]{article}
\usepackage{graphicx}
\usepackage{subcaption}

\begin{document}

\begin{figure}[h]
\centering

\begin{subfigure}{0.4\textwidth}
\centering
\includegraphics[width=\linewidth]{image1.png}
\caption{First Image}
\label{fig:img1}
\end{subfigure}
\hspace{10mm}
\begin{subfigure}{0.4\textwidth}
\centering
\includegraphics[width=\linewidth]{image2.png}
\caption{Second Image}
\label{fig:img2}
\end{subfigure}

\caption{Side by Side Images}
\label{fig:sidebyside}

\end{figure}

Reference: Figure \ref{fig:sidebyside}

\end{document}
```

Experiment No. 7

7. Develop a LaTeX script to create a document that consists of the following two mathematical equations

```
\documentclass{article}
\usepackage{amsmath}

\begin{document}

\section*{Equation 1}
\begin{align*}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
  &= \frac{-2 \pm \sqrt{4 + 32}}{2}
\end{align*}

\section*{Equation 2}
\begin{align*}
\varphi^\lambda_\sigma A_t
&= \sum_{\pi \in C_t} \text{sgn}(\pi)\varphi^\lambda_\sigma \varphi^\lambda_\pi \\
&= A_{\sigma t}\varphi^\lambda_\sigma
\end{align*}

\end{document}
```

Experiment No. 8

8. Develop a Latex script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

```
\documentclass{article}
\usepackage{amsthm}
\usepackage{amsmath}

\newtheorem{theorem}{Theorem}
\newtheorem{definition}{Definition}
\newtheorem{corollary}{Corollary}
\newtheorem{lemma}{Lemma}

\begin{document}

	\begin{theorem}
		In a right triangle,
		\[
		x^2 + y^2 = z^2
		\]
	\end{theorem}

	\begin{definition}[Absolute Value]
		\[
		|x|=
		\begin{cases}
			x & x \geq 0 \\
			-x & x < 0
		\end{cases}
		\]
	\end{definition}

	\begin{corollary}
		Hypotenuse is greater than each side.
	\end{corollary}

	\begin{lemma}
		$b = ra$
	\end{lemma}
```

\end{document}

Experiment No. 9

9: Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in
the section

NOTE: FILE NAME AS (main.tex)

```
\documentclass[12pt]{article}

\usepackage[backend=biber,style=ieee]{biblatex}

\addbibresource{references.bib}

\begin{document}

    \title{Service Placement in Mobile Edge Computing}
    \author{Student Name}
    \date{\today}

    \maketitle

    \section{Introduction}

    Mobile Edge Computing (MEC) has become an important technology for reducing latency and improving service delivery in next-generation networks. Researchers have proposed several service placement techniques to optimize resource utilization and user experience \cite{hardtoshare}. Multi-cell MEC architectures further enhance scalability and reduce communication overhead in distributed environments \cite{multicell}. Efficient bandwidth allocation and placement strategies are also essential for maintaining quality of service in MEC networks \cite{bandwidth}.


    \section{References}

    \printbibliography

\end{document}
```

NOTE: FILE NAME AS (references.bib)

```
@inproceedings{hardtoshare,
  author    = {He, Ting},
  title     = {It's Hard to Share: Joint Service Placement},
  booktitle = {IEEE ICDCS},
  year      = {2018},
  pages     = {365--375}
}

@inproceedings{multicell,
  author    = {Poularakis, Konstantinos},
  title     = {Joint Service Placement in MEC Networks},
  booktitle = {IEEE INFOCOM},
  year      = {2019},
  pages     = {10--18}
}

@article{bandwidth,
  author  = {Poularakis, Konstantinos},
  title   = {Service Placement in MEC Networks},
  journal = {IEEE/ACM Transactions on Networking},
  year    = {2020},
  volume  = {28},
  number  = {3},
  pages   = {1047--1060}
}

@article{edgeuav,
  author  = {Qu, Yuben},
  title   = {Service Provisioning for UAV MEC},
  journal = {IEEE JSAC},
  year    = {2021},
  volume  = {39},
  number  = {11},
  pages   = {3287--3305}
}

@article{edgeuncertainty,
```

```
  author  = {Xu, Xiaolong},
  title   = {Dynamic Resource Provisioning in Edge Computing},
  journal = {Concurrency and Computation},
  year    = {2020}
}
```

Experiment No. 10

10. Develop a Latex script to design a simple tree diagram or hierarchical
structure in the document with appropriate labels using the Tikz library

```
\documentclass[12pt]{article}

\usepackage{tikz}
\usetikzlibrary{trees}

\begin{document}

\title{Tree Diagram using TikZ}
\author{Student Name}
\date{\today}

\maketitle

\section{Hierarchical Structure}

\begin{center}
\begin{tikzpicture}[
  level 1/.style={sibling distance=50mm},
  level 2/.style={sibling distance=25mm},
  every node/.style={draw, rectangle, rounded corners, align=center}
]

\node {University}
    child {node {Engineering}
        child {node {CSE}}
        child {node {ECE}}
    }
    child {node {Management}
        child {node {MBA}}
        child {node {BBA}}
    };

\end{tikzpicture}
```

\end{center}

\end{document}

Experiment No. 11

11. Develop a LaTeX script to present algorithm in the document using algorithm2e library

```
\documentclass[12pt]{article}

\usepackage[ruled,vlined]{algorithm2e}

\begin{document}

\title{Algorithm Representation using algorithm2e}
\author{Student Name}
\date{\today}

\maketitle

\section{Bubble Sort Algorithm}

\begin{algorithm}[H]
\caption{Bubble Sort}
\KwIn{Array A of size n}
\KwOut{Sorted Array in Ascending Order}

\For{$i \leftarrow 0$ \KwTo $n-1$}{
   \For{$j \leftarrow 0$ \KwTo $n-i-2$}{
      \If{$A[j] > A[j+1]$}{
         Swap $A[j]$ and $A[j+1]$
      }
   }
}

\Return Sorted Array

\end{algorithm}

\end{document}
```

Experiment No. 12

12. Develop a LaTeX script to create a simple report and article by using suitable
commands and formats of user choice

Note : file name as(article.tex)

```
\documentclass[12pt]{article}

\usepackage{graphicx}

\title{Artificial Intelligence}
\author{Student Name}
\date{\today}

\begin{document}

\maketitle

\tableofcontents

\section{Introduction}
Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

\section{Applications}
AI is widely used in healthcare, education, robotics, banking, and autonomous vehicles.

\subsection{Healthcare}
AI helps doctors in disease prediction and medical diagnosis.

\subsection{Education}
AI-based systems provide smart learning and personalized education.

\section{Conclusion}
Artificial Intelligence is transforming modern technology and improving efficiency in various domains.

\end{document}
```

Note : file name as(report.tex)

```latex
\documentclass[12pt]{report}

\usepackage{graphicx}

\title{Mobile Edge Computing Report}
\author{Student Name}
\date{\today}

\begin{document}

\maketitle

\tableofcontents

\chapter{Introduction}
Mobile Edge Computing (MEC) provides computing resources near the users to reduce latency and improve performance.

\chapter{Architecture}
MEC architecture consists of edge servers, cloud servers, and user devices.

\section{Edge Servers}
Edge servers process user requests locally.

\section{Cloud Servers}
Cloud servers provide centralized storage and computation.

\chapter{Applications}
MEC is used in IoT, smart cities, healthcare, and autonomous vehicles.

\chapter{Conclusion}
MEC improves network efficiency and supports real-time applications.

\end{document}
```